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(*) Conclusion:- An outcome (or) decision, after considering all information (&) evidence.

→ Conclusion is same for all.

(*) Inference:- logical assumptions, drawn from available evidence, often serving as a tentative step in reasoning process. (or) Interpretation

→ Inference is different for all.

Involves using sample data, to draw a conclusion about entire data.

(II) Inferential stastics:- population :- over data

Sample Random choices from overall data

Point estimate :- Single value | Bias :- systematic error
Interval estimate :- Range (&) Difference | Variance :-

(*) Hypothesis:- (H_a (or) H_1):-

Specific prediction that a relationship (or) effect exist in between variables, representing the ~~Researcher~~ way to support their interpretation.

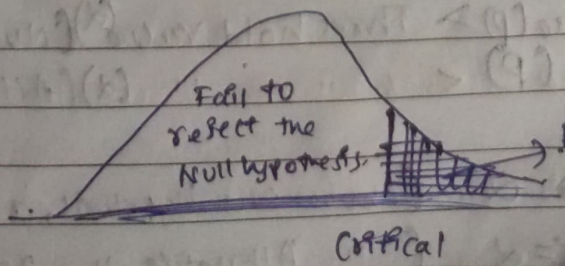
Types:

(*) Null Hypothesis (H_0):- Default Assumption that no relationship (or) effect exists blw variables.

(*) Alternative Hypothesis:- (H_1 (or) H_a):- The statement that contradicts the null, proposing that there is an effect (or) relationship blw variables.

(*) one tailed test:- A stastical test that evaluates ~~that~~ the possibility of a relationship in only one direction, either greater than (or) less than, with entire rejection place in one tail of the distribution.

Eg:

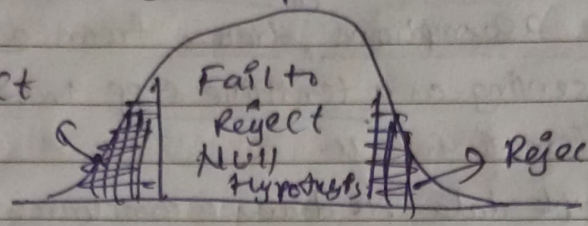


Null Hypothesis :- Assumption we made

Alternative Hypothesis :- opposite of Null hypothesis.

Eg :- Null Hypothesis :- girls average marks > 600 ; Alternative hypothesis :- Opposite (girls average marks < 600).

(*) Two Tailed test :- A statistical test, evaluates possibility of relationship in both directions, with rejection region split b/w both tails of the distribution.



(*) Critical Value :- The boundary values that define the rejection regions, if the test exceeds this value, the null hypothesis is rejected.

Hypothesis Working Flow :-

1. Define Ho and H1. (state Null & Alternative hypothesis clearly).
2. Choose test statistic. (Pick z-test, t-test (or) Chi-square test based on data)
   - when sample size is less :- t-test (sample size < 30)
   - " " " more :- z-test (samplesize >= 30)
   - when categorical data → Chi-square test.
   - when we know population variants → z-test
   - if we don't know " " → t-test.

   Variants :- deviation from mean.

   Task :- z-test, t-test, Chi-square test) with formulas, working. (definition, example)
3. Select Significance Level (α) (Threshold Value)
   - Select threshold (commonly 0.05) for rejecting Ho.
   - we get, statistical probability value (we get from the z-test, t-test, chi-square test).
   - statistical probability value (p) > Threshold value (α) (Null hypothesis).
   - " " " (p) < " " (α) (Alternative hypothesis)
4. Choose decision :- $p > \alpha$ (choose Null hypothesis, reject Alternative)
   $p <= \alpha$ (" Alternative " , reject Null hypo)

Just like ~~A~~ Confusion matrix:-

	Failed to reject Null Hypothesis (Accept Null Hypo)	Reject Null Hypothesis
Null Hypothesis True (Ho True)	TRUE correct decision $(1-\alpha)$	Type I Error (α)
Null Hypo is False (Ho False)	Type II Error (β)	TRUE correct Decision power $(1-\beta)$

Applications of Inference testing:-

A/B testing, Feature selection, Clinical AI, Recommendation Systems

① Z-TEST:- Z-Test is used to determine whether a sample result is significantly different from a hypothesis population value

When to use:-

① Numerical data
② Large samples → $n (>= 30)$
③ Standard deviation (σ) of data (or) population is known.
④ Also, used for proportions.

Type:- ①:- One-Sample Z-Test:- used for comparing Sample mean (vs) population mean

Formulae; $$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$ where; $\bar{X}$ = Sample mean, μ = population mean, σ = population Standard deviation, n = Sample size

Eg: Company says, average battery life = 50 hours
we test 36 batteries. (n)

given; population mean (μ) = 50
Sample mean ($\bar{x}$) = 53 (avg of 36 batteries)
population S.D (σ) = 9
Sample size (n) = 36

By Formulae: $$Z = \frac{X - \mu}{\sigma / \sqrt{n}} = \frac{53-50}{9/\sqrt{36}} = \frac{3}{9/6} = \frac{3}{1.5} = 2;$$

$Z = 2$

Eg ②:- H_0 = null hypothesis (girl's average score less than 600 in exam).

H_1 = Alternative " (" " " " greater " " ").

given, S.D of girl's is 100 (S.D = 100)

size (n) = 20 ; population mean (μ) = 600.

Samples: 650, 730, 510, 670, 480 - - -

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(2) Two Sample z-test:- we perform, Two sample z-test, when we want to compare the mean of two sample.

$$z\text{-score} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Difference blw Sample mean

Diff blw population mean $(\mu_1 - \mu_2)$.

σ_1, σ_2 → S.D of 1, 2 ||| n_1, n_2 → sample sizes

Eg: Sample mean of Girls $(\bar{x}_1) = 641$
" " " Boys $(\bar{x}_2) = 613.3$
S.D of Population of Girls $(\sigma_1) = 100$
" " " " Boys $(\sigma_2) = 90$

Sample size of both = 20.
$n_1 = 20, n_2 = 20$
① Diff blw mean of popution $(\mu_1 - \mu_2) = 20$.

$$\Rightarrow \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \Rightarrow \frac{(641 - 613.3) - 10}{\sqrt{\frac{100^2}{20} + \frac{90^2}{20}}} = 0.588$$

z-score = 0.588

So; $P(z \geq 0.588) \approx 0.278$.

P Value = 0.278

⇒ critical value = 1.645
⇒ z-score < critical value
p-value > 0.05 ⇒ we fail to to reject Null hypothesis
⇒ So, we accept H_0.

(3) z-test for proportion:-

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$\hat{p}$ = Sample proportion.
p_0 = Hypothesized Population proportion.
n = Sample size.

(II.) T-Test:- T-test is used to determine whether there is a significant difference blw mean, when population standard deviations are not known.

① Used for smaller Samples.

Type-1: One-sample T-test:

$$t = \frac{\bar{X} - M}{S/\sqrt{n}}$$

where; $\bar{x}$ = Sample mean
μ = Hypothesized mean
S = Sample SD.
n = Sample size.

Degress of freedom ⇒ $df = n - 1$

Eg: Teacher claims Avg. student score = 70; we have 16 students.

given; Sample mean = 74
Hypothesized mean = 70
Sample SD = 8
n = 16

$$t = \frac{74 - 70}{8/\sqrt{16}}$$

$$t = \frac{4^2}{2^8/4} = 2 \Rightarrow t = 2$$

∴ Degrees of freedom ⇒ $df = 16 - 1 = 15$

For, two-tailed test at $\alpha = 0.05$: $t_{critical} \simeq 2.131$

since;

$2 < 2.131$ Reject H_0.

Type-2: Independent Two-Sample T-test:-

To Compare Mean of Group A vs Mean of Group B. Two groups should be Independent.

Eg:

Class A	Class B
mean = 75	mean = 70
SD = 8	SD = 7
n = 2	n = 20

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

⇒ Substitute; $t = \frac{75 - 70}{\sqrt{\frac{8^2}{2} + \frac{7^2}{20}}}$

$t \approx 2.10$

If $P < 0.05$: Reject H_0
$P > 0.05$; Reject H_1

(III.) CHI-SQUARE TEST:- mainly used for categorical data (or) frequency (or) count data.

↳ Categories + Counts → Chi-square

Type 1:- Chi-square Goodness-of-Fit:- Observed frequencies (VS) Expected frequencies

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

O = observed frequency.
E = Expected frequency.

Eg: A shop expects customer to choose four products equally.

product	OBSERVED	Expected
A	30	25
B	20	25
C	25	25
D	25	25

Calculate each distribution:-

For A: $\frac{(30-25)^2}{25} = 1$ (E)

For B: $\frac{(20-25)^2}{25} = 1$

For C: $\frac{(25-25)^2}{25} = 0$

For D: $\frac{(25-25)^2}{25} = 0$

$\chi^2 = 1+1+0+0$; $\boxed{\chi^2 = 2}$

Degree of freedom ; $df = k-1$ (or) $df = n-1$ (n, k = no. of samples)

$\boxed{df = 4-1 = 3}$

At $\alpha = 0.05$:

$df = 3$

$\boxed{\chi^2_{critical} \approx 7.815}$ (From chi-square distribution table, Pre defined table; we check values directly from it)

$2 < 7.815$

→ Failed to reject H_0

Conclu: The observed product choices are not significantly different from an equal distribution.

**** Master Rule:-

Calculated Value crosses the critical boundary

↓

Reject H_0 ✓

If not; we use; Failed to Reject H_0 ✓.

Q: ① Z-Test: Did girls score more than 600?

given; Z-calculated = 1.83

Z critical = 1.645 (Right tailed, $\alpha = 0.05$)

since,

$1.83 > 1.645$ ✓

✓ Reject H_0

Conclusion: Girls score more than 600. (significancy)

② T-test:- Is average score different from 70?

Given; t calculated = 2.30

t critical = 2.13

since; $2.30 > 2.13$ Reject H_0 (✓)

→ Average score is significantly different from 70.

③ Chi-square test:- Are product choices equally distributed?

Given; X^2 calculated = 2

X^2 critical = 7.815

Since; $2 < 7.815$ Failed to reject H_0 (X)

Conclusion: Not enough evidence to say product choices are different.

2 methods

Calculate Value > Critical Value (✓ Reject H_0)

" " < " (X Fail to Reject H_0)

P-Value method:-

P-Value $\leq \alpha$ → Reject H_0

P-Value $\geq \alpha$ → Fail to Reject H_0

Type-2:- Chi-Square Test for Independence:

→ Two categorical Variables are related / associated.

Eg: To know gender associated with Product Preference.

Data:

	Product A	Product B	Total
Male	40	20	60
Female	20	40	60
Total	60	60	120

Step:-1 Calculate Expected frequency

$$E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

For male & Product A; $E = \frac{60 \times 60}{120} \Rightarrow E = 30$

Similar for all cases. (Male & Female + A & B)

Expected table:-

	Product A	Product B
Male	30	30
Female	30	30

Step-2: Apply chi-square Formula:-

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

For four cells: $\chi^2 = \frac{(40-30)^2}{30} + \frac{(20-30)^2}{30} + \frac{(20-30)^2}{30} + \frac{(40-30)^2}{30}$

$$\chi^2 = 13.33$$

Degree of freedom $\Rightarrow$ $df = (r-1)(c-1)$

$df = (2-1)(2-1) = 1$

At $\alpha = 0.05$: $\chi^2_{Critical} = 3.841$

Since, $13.33 > 3.841$

Reject H_0

Conclusion:- gender (&) Product are statistically associated.